Appendix of "TempFlash: GPU-Resident, Single-Launch Training of Temporal Graph Neural Networks"

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This appendix accompanies the paper. References to Sections 1–8, Tables 1 and 2, Figures 1–8 and Equations (1)–(7) point to the paper; the bibliography for this document appears at its end.

A PROOF OF PROPOSITION 1

Reference semantics. Consider TGL's recent sampler with a single history and layer, the configuration used here. Node v's non-decreasing timestamps occupy T-CSR segment $[o_v, o_{v+1})$, and reset initializes p_v to o_v. For roots (v_r, t_r), $r = 1, \ldots, R$, with fp32 thresholds $\theta_r = \mathrm{fl}(t_r - 10^{-7})$, all pointer updates precede all root sampling. Updating root r scans $j = p_{v_r}, p_{v_r} + 1, \ldots$ to the first j satisfying $ts_j > \theta_r$; when $j \neq p_{v_r}$ it assigns $p_{v_r} \leftarrow j - 1$, otherwise retains the pointer, or assigns $o_{v_r+1} - 1$ upon reaching the segment's end. A per-node lock serializes updates in an arbitrary order. To sample root r, entries $k = p_{v_r}, p_{v_r} - 1, \ldots$ are visited down to but not including $\max(o_{v_r}, p_{v_r} - K)$; those satisfying $ts_k < \theta_r$ are emitted in visitation order, from newest to oldest.

Lemma 1 (one update). *Define u_r as the first index in v_r's segment with timestamp greater than θ_r, or o_{v_r+1} if no such entry exists. Updating root r replaces p_{v_r} by* $\max(p_{v_r}, u_r - 1)$. Since the segment is sorted, timestamps exceeding θ_r occupy its suffix $[u_r, o_{v_r+1})$. For $u_r > p_{v_r}$, scanning either stops at $j = u_r \neq p_{v_r}$ or reaches the segment boundary when $u_r = o_{v_r+1}$, giving $u_r - 1$ in both cases. For $u_r \leq p_{v_r}$, the first entry stops the scan without changing the pointer; an empty segment performs no scan, with $u_r - 1 < o_{v_r} = p_{v_r}$. Thus all cases yield $\max(p_{v_r}, u_r - 1)$.

Lemma 2 (order independence). *Regardless of update order, the batch leaves* $p_v = \max\left(p_v^{\mathrm{old}}, \max_{r:v_r=v}(u_r - 1)\right)$. Lemma 1 expresses each update as $x \mapsto \max(x, c_r)$; these maps commute, and composing them gives $x \mapsto \max(x, \max_r c_r)$, precisely the batch pointer rule in Eq. (4).

Lemma 3 (search). *Searching the key array for the upper bound of $(v_r \ll 32) \mid \varphi(\theta_r)$ returns u_r.* Equation (2)'s φ follows floating-point ordering for non-NaN fp32 values apart from putting -0 below $+0$; neither timestamps shifted to start at zero nor thresholds from fp32 subtraction take the value -0. Consequently, for node v_r's entry i, the condition that κ_i exceeds the query is equivalent to $ts_i > \theta_r$, while other nodes' entries compare according to node identifier. The first greater key in this sorted array is therefore at u_r.

Proof of Proposition 1. The GPU computes θ_r using the identical fp32 subtraction. Applying Lemmas 2 and 3 inductively over batches after reset shows that maximum-reduction scatter produces the same pointers as the reference after each update phase. With $p = p_{v_r}$ and $m = \min(p - o_{v_r}, K)$, the reference emits entries $p - j$ for $j = 0, \ldots, m - 1$ whose timestamps lie below θ_r, preserving that order. GPU slot j accesses $p - j$ and applies exactly these conditions, so reading valid slots rowwise recovers the reference sequence. Both read neighbor identifiers, event identifiers and timestamps from identical entries and calculate deltas through the same fp32 subtraction. □

Table 3: State commits: frequency of retaining an update other than the final occurrence (CPU: counts; GPU and Write: percentage of repeated nodes).

Data	Model	Seq.	Batches	Updates	Dup.	CPU	GPU	Write
LastFM	TGN	mails	768	1.81M	296.2K	0	55.9%	57.1%
LastFM	TGN	memories	768	1.81M	296.2K	0	57.9%	58.6%
LastFM	APAN	mails	768	11.70M	678.0K	0	80.8%	81.2%
Wiki-Talk	TGN	mails	2,589	10.97M	1.64M	0	66.3%	67.0%
Wiki-Talk	TGN	memories	2,589	10.97M	1.64M	0	57.4%	58.5%
Wiki-Talk	APAN	mails	2,589	65.34M	6.33M	0	77.9%	78.1%
Stack-Ovf.	TGN	mails	300	1.49M	243.1K	0	61.8%	62.3%
Stack-Ovf.	TGN	memories	300	1.49M	243.1K	0	63.8%	64.5%
GDELT	TGN	mails	300	1.58M	148.4K	0	79.7%	80.0%
GDELT	TGN	memories	300	1.58M	148.4K	0	80.1%	80.6%

B CORRECTNESS TESTS

Sampler. In addition to the four Table 1 graphs (§5.3), we tested the GPU sampler against TGL's C++ sampler on 2,000 random graphs designed to frequently exercise corner cases: few distinct times, many ties, repeated roots, nodes with no or little prior history, and thresholds equal to stored timestamps. Each graph undergoes two passes separated by a reset, using $K \in \{1, 2, 3, 10\}$ and either one or eight sampler threads. The 41,404 batches contain 1.12M roots and 963.6K sampled neighbors, including 397,571 threshold-equal entries, 38,789 repeated-root batches, 248,918 excluded first entries, 16,752 empty histories and 431,282 short histories; across them, neighbor, event-identifier, timestamp and ordering differences total 0.

Commit semantics. We constructed each batch's candidate sequence in reference order (TGN: event-interleaved mails, source memories followed by destination memories; APAN: root mails followed by mails to each root's sampled neighbors) and checked last-occurrence agreement for ETC selection on CPU, selection on GPU, and plain GPU indexed writes without selection. Table 3 records, per sequence, inspected batches (300 strided across Stack-Overflow and GDELT epochs), update counts and nodes repeated within a batch, summed across batches. Sequential CPU selection always retains the last occurrence, while GPU selection keeps a different update for 56–81% of repeated nodes, roughly as often as unselected indexed writes. Repeating GPU execution on identical batches can also change the outcome. By construction, TempFlash always commits each node's final occurrence (§3.5).

Table 4: Training time per epoch (s) for the TGAT and APAN ports of Orca and Zebra, alongside TempFlash (Table 2).

	LastFM		Wiki-Talk		Stack-Overflow		GDELT	
	TGAT	APAN	TGAT	APAN	TGAT	APAN	TGAT	APAN
Orca	9.91	14.5	72.1	228.2	1390.7	2006.8	5095.3	1244.6
Zebra	36.6	41.6	212.4	418.5	2942.1	3208.6	9837.5	4900.3
TempFlash	1.64	3.49	13.2	15.7	126.0	159.2	330.8	347.3

Differential test. We initialize TempFlash from an ETC model instance and train it alongside ETC with identical features, adaptive batches and negative destinations, disabling dropout. After each batch, comparisons cover relative prediction and loss differences, every post-Adam parameter, absolute differences in touched-node memories and mails, and exact equality of memory timestamps, mail timestamps and ring pointers. CPU selection makes the reference retain the final occurrence (Table 3). Across all 768 LastFM training batches, TGN and APAN discrete states match exactly, with 0 differing timestamps or pointers. The released TGN reference instead selects on the GPU and differs in 187,586 memory timestamps and 163,568 mail timestamps across those 768 batches; GPU selection for APAN yields 242,221 and 764,260 differences, respectively. Re-synchronizing TempFlash to reference state before each batch bounds maximum relative differences across 300 batches of all three models at 1.6×10^{-6} for predictions and 2.4×10^{-6} for parameters.

Rounding against fp64. An fp64 reference-model copy evaluates identical neighborhoods and pre-step states for 6 synchronized TGN batches. Median relative fp32 prediction errors are 3.5×10^{-7} for TempFlash and 4.6×10^{-7} for the reference; maximum relative gradient errors are 8.8×10^{-7} and 2.2×10^{-5}, respectively, so TempFlash introduces no greater rounding error.

C BASELINES

Orca and Zebra. Their releases build on TGN's implementation [1] and support TGN alone. We added ports for the other two models. TGAT is obtained as the memory-free case: removing node memory and using node features at layer 0 while retaining two attention layers, Orca's reused embeddings and Zebra's PageRank neighborhoods. Our APAN port follows its default configuration: ten mails per node, TGL's attention memory updater with layer normalization for embeddings, and mail delivery to each node and its system-sampled neighbors (Orca's ten most recent, or Zebra's top-k). Port epoch times in Table 4 are 3.6–29.7 times TempFlash's.

D ADDITIONAL DYNAMIC GRAPHS

Accuracy and speed. Table 5 reports three-seed test AP and epoch duration for TempFlash and ETC on the 36 workloads introduced in §5.8. Their average AP gap is 0.69 points; on 35 workloads it remains within twice the larger seed standard deviation, or one point. Back-to-back timing against ETC on a dedicated GPU gives TempFlash a 3.7–8.3× per-epoch speedup across all 36 workloads (geometric mean 5.2×).

Algorithm 1 Training step captured for size bucket B

Require: cursor c; batch starts $lo[\cdot]$ and sizes $sz[\cdot]$; event arrays; T-CSR; tiers; states

$n \leftarrow sz[c]$; $ok \leftarrow [0, B) < n$; $ev \leftarrow \min(lo[c] + [0, B), |E| - 1)$ ▷ device cursor

$(\mathbf{src}, \mathbf{dst}, \mathbf{t}) \leftarrow \text{events}[ev]$; $\mathbf{neg} \leftarrow \text{randint}(|V|, B)$

roots $\leftarrow [\mathbf{src}; \mathbf{dst}; \mathbf{neg}]$ at times $[\mathbf{t}; \mathbf{t}; \mathbf{t}]$

$(nbr, eid, \Delta t, mask) \leftarrow \textsc{Sample}(\text{roots}, ok)$ ▷ §3.2

read features, memories and mails into slots ▷ §3.4, §3.6

$\mathcal{L} \leftarrow \textsc{MaskedBCE}(\textsc{Model}(\cdot), ok)$; $\nabla\mathcal{L}$; Adam step ▷ §3.3

Commit(new memories and mails, ok) ▷ §3.5

$c \leftarrow c + 1$

E IMPLEMENTATION DETAILS

Sampling time. Table 7 measures one epoch's training-batch sampling with TGL's C++ implementation on 1, 8 and 32 host threads and TempFlash's GPU sampler using either individual launches (eager) or captured replay. Identical roots produce identical neighbors in both samplers; Nbrs. lists the epoch's total neighbor count. Captured GPU sampling outpaces eight-thread TGL by 9.9–17× and 32-thread TGL by 10–29×.

Memory. Table 6 accounts for TempFlash GPU storage by T-CSR (Entries gives its entry count; T-CSR includes sorting keys), event arrays, node and edge features, node states (memories and mailboxes), and remaining tensors including step workspace; Host records the edge-feature table held in pinned, zero-copy host memory.

Step anatomy. With all data resident, a training step issues 344–485 kernels (Table 8); captured replay avoids their individual host dispatches and improves step time by up to 1.9×, with the greatest benefit for small-batch memory-based models.

Table 5: Results on additional graphs: seed counts, test AP (%, mean ± standard deviation), epoch time (s), and speedup.

Data	Model	n	ETC AP	TempFlash AP	ETC	TempFlash	Spd.
Wikipedia	TGN	3	97.53 ± 0.27	97.30 ± 0.19	3.14	0.84	3.8×
	TGAT	3	88.64 ± 0.21	88.62 ± 0.25	2.46	0.37	6.6×
	APAN	3	91.07 ± 0.65	91.01 ± 0.27	2.87	0.50	5.7×
Reddit	TGN	3	99.14 ± 0.63	99.50 ± 0.05	18.1	3.72	4.9×
	TGAT	3	96.36 ± 0.25	96.09 ± 0.12	7.86	1.45	5.4×
	APAN	3	98.52 ± 0.60	98.75 ± 0.10	12.1	1.89	6.4×
MOOC	TGN	3	99.34 ± 0.04	99.37 ± 0.04	6.78	1.69	4.0×
	TGAT	3	97.13 ± 0.08	97.14 ± 0.02	5.17	0.72	7.1×
	APAN	3	98.87 ± 0.06	98.89 ± 0.09	5.39	0.98	5.5×
Enron	TGN	3	71.88 ± 2.02	70.84 ± 1.97	2.73	0.55	4.9×
	TGAT	3	70.75 ± 0.67	70.65 ± 0.40	1.95	0.31	6.3×
	APAN	3	52.99 ± 1.54	53.58 ± 1.96	1.85	0.33	5.6×
SocialEvo	TGN	3	80.56 ± 0.05	82.46 ± 0.31	38.1	7.69	5.0×
	TGAT	3	77.94 ± 0.07	77.90 ± 0.02	23.4	3.52	6.7×
	APAN	3	63.96 ± 4.86	60.77 ± 9.76	23.9	4.32	5.5×
UCI	TGN	3	80.15 ± 3.91	76.86 ± 7.22	1.29	0.33	3.9×
	TGAT	3	63.11 ± 1.25	63.17 ± 0.17	0.97	0.20	4.8×
	APAN	3	62.61 ± 2.17	62.47 ± 1.13	1.08	0.21	5.1×

Data	Model	n	ETC AP	TempFlash AP	ETC	TempFlash	Spd.
Flights	TGN	3	90.38 ± 1.63	90.85 ± 1.66	32.5	8.43	3.9×
	TGAT	3	82.12 ± 0.08	82.08 ± 0.06	19.6	3.33	5.9×
	APAN	3	87.35 ± 0.36	88.01 ± 1.46	24.0	4.43	5.4×
CanParl	TGN	3	72.09 ± 3.34	72.85 ± 1.44	3.06	0.37	8.3×
	TGAT	3	71.18 ± 0.22	71.26 ± 0.24	0.93	0.19	5.0×
	APAN	3	63.41 ± 0.67	64.76 ± 2.09	0.98	0.22	4.5×
USLegis	TGN	3	74.58 ± 0.68	75.88 ± 0.29	1.29	0.25	5.1×
	TGAT	3	68.08 ± 0.33	68.05 ± 0.59	0.85	0.13	6.4×
	APAN	3	65.66 ± 4.88	64.16 ± 3.98	0.81	0.16	5.1×
UNtrade	TGN	3	65.43 ± 0.59	65.58 ± 0.86	7.46	1.89	4.0×
	TGAT	3	53.37 ± 0.05	53.38 ± 0.04	5.63	0.87	6.5×
	APAN	3	60.01 ± 1.62	60.90 ± 3.32	5.90	1.09	5.4×
UNvote	TGN	3	58.56 ± 1.41	57.96 ± 0.35	14.1	3.84	3.7×
	TGAT	3	50.93 ± 0.01	50.95 ± 0.03	11.0	1.71	6.4×
	APAN	3	55.60 ± 1.27	55.28 ± 1.10	10.9	2.18	5.0×
Contacts	TGN	3	89.97 ± 0.17	89.56 ± 1.12	42.1	10.4	4.1×
	TGAT	3	88.45 ± 0.08	88.38 ± 0.10	28.0	4.59	6.1×
	APAN	3	87.05 ± 1.43	82.48 ± 5.29	24.7	5.83	4.2×

Table 6: Array-level GPU allocation (GB) for each TempFlash workload, together with edge features in zero-copy host storage. ^{p} denotes bit packing; ^{f} denotes mailbox factorization.

Data	Model	Entries	T-CSR	Events	Node feat.	Edge feat.	States	Other	Total	Host
LastFM	TGN	2.59M	0.05	0.01	0.00	0.62	0.00	0.12	0.80	0.0
LastFM	TGAT	2.59M	0.05	0.01	0.00	0.62	0.00	0.09	0.77	0.0
LastFM	APAN	2.59M	0.05	0.01	0.00	0.62	0.03	0.10	0.80	0.0
Wiki-Talk	TGN	7.83M	0.16	0.09	0.73	5.02	2.03	0.17	8.20	0.0
Wiki-Talk	TGAT	7.83M	0.16	0.09	0.73	5.02	0.00	0.13	6.13	0.0
Wiki-Talk	APAN	7.83M	0.16	0.09	0.00	5.02	9.07^{f}	0.13	14.47	0.0
Stack-Ovf.	TGN	63.50M	1.22	0.71	1.67	0.00	4.63	0.27	8.50	40.7
Stack-Ovf.	TGAT	63.50M	1.22	0.71	1.67	0.00	0.00	0.21	3.81	40.7
Stack-Ovf.	APAN	63.50M	1.22	0.71	0.00	0.00	20.70^{f}	0.22	22.85	40.7
GDELT	TGN	191.29M	3.56	2.14	0.03	4.10^{p}	0.03	0.39	10.24	0.0
GDELT	TGAT	191.29M	3.56	2.14	0.03	4.10^{p}	0.00	0.37	10.19	0.0
GDELT	APAN	191.29M	3.56	2.14	0.03	4.10^{p}	0.24	0.35	10.42	0.0

Table 7: Sampling duration (s) for one epoch's training batches using TGL's C++ implementation or TempFlash's GPU sampler. Nbrs. counts neighbors sampled per epoch.

		TGL sampler (threads)			TempFlash sampler	
Data	Nbrs.	1	8	32	eager	captured
LastFM	16.2M	1.5	1.2	1.2	0.58	0.12
Wiki-Talk	57.6M	10.3	5.4	5.7	2.08	0.50
Stack-Ovf.	673.6M	99.1	49.7	49.6	11.64	3.25
GDELT	1586.6M	165.3	119.9	198.8	30.41	6.93

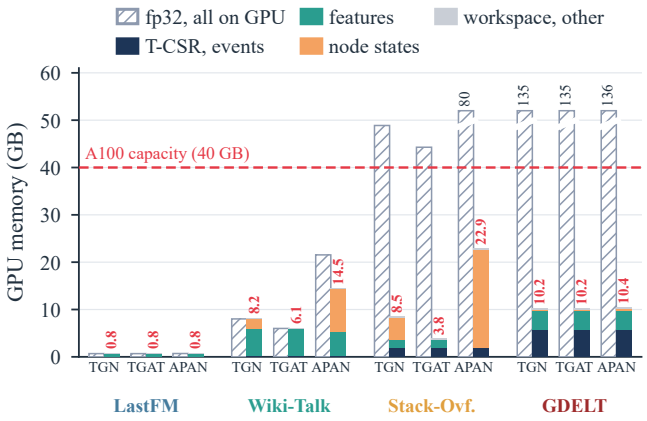


Figure 9: Per-workload GPU footprint under full fp32 storage (hatched) and TempFlash allocation (solid).

Table 8: Kernel counts and wall-clock step times with default settings and fully resident data: individual kernel launches (Eager) versus captured replay (Replay).

Data	Model	Kernels	Eager (ms)	Replay (ms)	Ratio
LastFM	TGN	474	8.0	6.0	1.3×
	TGAT	349	10.5	9.7	1.1×
	APAN	377	6.6	3.4	1.9×
Wiki-Talk	TGN	468	13.0	12.4	1.0×
	TGAT	345	15.0	14.5	1.0×
	APAN	392	6.7	6.1	1.1×
Stack-Ovf.	TGN	470	17.9	17.3	1.0×
	TGAT	344	19.1	18.6	1.0×
	APAN	393	10.0	9.4	1.1×
GDELT	TGN	485	13.4	12.7	1.1×
	TGAT	350	11.5	11.0	1.0×
	APAN	379	7.6	7.0	1.1×

REFERENCES

[1] Emanuele Rossi, Ben Chamberlain, Fabrizio Frasca, Davide Eynard, Federico Monti, and Michael Bronstein. 2020. Temporal Graph Networks for Deep Learning on Dynamic Graphs. *arXiv preprint arXiv:2006.10637* (2020).